
Chapter 1: Proof

A Level Pure Maths Notes · Rexponential

Why this chapter matters

Proof is new territory if you've come from a curriculum where you calculated and verified far more than you argued. At A level, a “show that” or “prove that” question is not asking you to check a fact is true for one number — it is asking you to show it is true for every number the question specifies, using logic alone.

This matters for two reasons. First, proof questions appear on every paper, often early, and they are usually worth more marks than students expect for how short they look. Second, and more importantly: proof is where you first meet the idea that mathematics is a chain of justified steps, not a sequence of answers. Get this chapter right with them, and the habit of justifying every line carries into every other topic.

This chapter also sets the pattern for the rest of the course: from Chapter 2 (Algebra and Functions) onward, every “show that” or “prove” question leans on the same deductive habit — represent the general case, manipulate it with full working, and conclude — so it is worth being fluent in this chapter before moving on.

How to use these notes

Each section below is split into four parts:

- **Concept** — what the idea means and why it works, in plain language before any symbols.
- **Procedure** — the exact steps to follow when answering a question of this type.
- **Worked example** — a full question with a step-by-step solution, followed by the marks a UK exam board would likely award.
- **Exam tip** — a common mistake or subtlety to watch for with this method, usually drawn from where students actually lose marks.

How this document is organised

These notes cover the ten chapters of the A Level Pure Maths specification, in the order they are usually taught: **1 Proof** (the logical toolkit used throughout), **2 Algebra and Functions**, **3 Coordinate Geometry**, **4 Sequences and Series**, **5 Trigonometry**, **6 Exponentials and Logarithms**, **7 Differentiation**, **8 Integration**, **9 Numerical Methods**, and **10 Vectors**. Chapters 7 and 8 (calculus) draw on almost every chapter before them, and Chapter 9 and the mechanics in Chapter 10 draw directly on Chapter 7 — so the chapters build cumulatively rather than standing alone.

There are four proof techniques on the A level Pure specification: **deduction**, **exhaustion**, **counterexample** (for disproof), and **contradiction**. These are easy to confuse, so as you go through, keep coming back to the one-line distinction: deduction builds a general argument; exhaustion checks every possible case; counterexample kills a claim with a single failing case; contradiction assumes the opposite of what you want and shows that assumption breaks down.

1.1 What a mathematical proof is

Concept

A proof is a logically complete argument that a statement is true in every case it claims to cover, with no gaps and no reliance on examples alone. This is a genuinely different standard from what you may be used to: checking that a formula works for $x = 1, 2$ and 3 feels convincing, but it proves nothing about $x = 4$, let alone every real number.

The key vocabulary you need before you can write a proof:

KEY VOCABULARY

- **Statement** — the claim to be proved (e.g. “the sum of two odd numbers is even”).
- **Given / assume** — what you are allowed to take as true at the start (definitions, given conditions).
- **Conjecture** — a statement believed to be true but not yet proved.
- **Theorem** — a statement that has been proved.

A useful way to think about this: try to “prove” that every number you pick, doubled, is even — then ask yourself how many numbers you’d need to try before you could be certain. The answer (infinitely many) is exactly why checking examples can never be a proof of a general statement, and exactly why algebra — which represents *any* number at once — is the tool that closes the gap.

Procedure: general structure of any proof

1. **State what you are proving** and, if useful, restate it in your own words or algebraic form.
2. **Represent the general case algebraically.** Do not use a specific number; use a letter (or letters) that could stand for any valid value.
3. **Manipulate algebraically**, showing every step, until you reach a form that clearly matches what you were asked to show.
4. **State the conclusion**, linking back to the original claim. A proof without a final concluding sentence often loses the last mark.

Worked example

EXAMPLE 1.1 (3 MARKS)

Prove that the sum of any two even numbers is even.

Step 1 Let the two even numbers be represented algebraically. Any even number can be written as $2m$ for an integer m , so let the two numbers be $2m$ and $2n$, where m and n are integers. (M1 — correct general representation)

Step 2 Add them: $2m + 2n = 2(m + n)$. (M1 — correct algebraic manipulation)

Answer Since m and n are integers, $m + n$ is also an integer. So $2(m + n)$ is 2 multiplied by an integer, which is by definition even. (A1 — conclusion linked to the definition of even)

EXAM TIP

The most common mark lost here is using $2m$ and $2m + 2$ for “two even numbers” — that only proves the claim for *consecutive* even numbers, not any two. Emphasise that different letters (m and n) are needed unless the question specifically says the numbers are related.

1.2 Proof by deduction

Concept

Deduction is the default proof method: starting from things known to be true (definitions, given facts, previously proved results) and reasoning forward, step by step, until the statement is established. Every step must follow logically and inevitably from the one before it — there is no room for “it looks like” or “it's probably”.

It is tempting to think deduction means “using algebra”, but the real definition is about the *logical structure*: each line is a necessary consequence of the previous lines and the given facts, not merely a re-arrangement that happens to be true. In practice, at A level, deduction usually does mean algebraic manipulation — but thinking of it as logical necessity shows why skipping steps, or writing a line that doesn't follow, breaks the proof even if the final answer is correct.

Procedure

1. Write down what is given or known (definitions of even/odd, the general form of an integer, etc.).
2. Represent the general case using algebra — letters, not specific numbers.
3. Perform algebraic steps, each one justified by ordinary algebra rules, moving from the given information towards the statement to be proved.
4. Interpret the final algebraic form in words, connecting it explicitly to what was asked.
5. Write a concluding sentence.

Worked example

EXAMPLE 1.2A (4 MARKS)

Prove that $(n + 1)^2 - (n - 1)^2$ is a multiple of 4 for every integer n .

Step 1 Expand each bracket: $(n + 1)^2 = n^2 + 2n + 1$ and $(n - 1)^2 = n^2 - 2n + 1$. (M1)

Step 2 Subtract: $(n^2 + 2n + 1) - (n^2 - 2n + 1) = 4n$. (M1 — correct simplification)

Step 3 Since n is an integer, $4n$ is 4 multiplied by an integer. (A1 — correct interpretation)

Answer Therefore $(n + 1)^2 - (n - 1)^2$ is always a multiple of 4, for every integer n . (A1 — conclusion, referring back to “every integer”)

EXAMPLE 1.2B (4 MARKS)

Prove that $1/n - 1/(n + 1) = 1/(n(n + 1))$ for all positive integers n .

Step 1 Start from the left-hand side and combine over a common denominator: $1/n - 1/(n + 1) = (n + 1)/(n(n + 1)) - n/(n(n + 1))$. (M1)

Step 2 Combine the numerators: $= [(n + 1) - n] / (n(n + 1))$. (M1)

Step 3 Simplify the numerator: $(n + 1) - n = 1$, so the expression becomes $1/(n(n + 1))$. (A1)

Answer This is exactly the right-hand side, so the identity is proved. (A1)

EXAM TIP

When a question gives an identity to prove (like example 2), start from the more complicated side and work towards the simpler side — never manipulate both sides at once or “cross-multiply”, since that assumes what you’re trying to prove. This is a very common error worth drilling early.

1.3 Proof by exhaustion

Concept

Proof by exhaustion works by breaking a statement down into a finite number of cases that between them cover every possibility, then checking each case is true. It is only usable when the total number of cases is finite (or can be reduced to a finite number, as in the example below) — you cannot exhaust the infinite set of all integers one at a time, which is exactly why deduction exists for those situations.

The method can feel less elegant than deduction, and that’s worth saying explicitly: exhaustion is a brute-force method, valid precisely because it leaves nothing out, but it only works when “everything” is a manageable, listable set.

Procedure

1. Identify a way to split all possible values into a small number of distinct cases that together cover every possibility (this is usually the hardest step — the cases must be exhaustive, meaning nothing is left out, and usually should not overlap).
2. Check the statement holds in each case in turn, showing the working for each.
3. State that since the statement holds in every case, and the cases cover every possibility, the statement is proved.

Worked example

EXAMPLE 1.3A (5 MARKS)

Prove that $n^2 + n$ is even for all integers n .

Step 1 Every integer n is either even or odd — these two cases cover all integers. (B1 — identifying the cases)

Step 2 Case 1: n is even. Write $n = 2k$ for some integer k . Then $n^2 + n = (2k)^2 + 2k = 4k^2 + 2k = 2(2k^2 + k)$, which is even since it is 2 times an integer. (M1 A1)

Step 3 Case 2: n is odd. Write $n = 2k + 1$ for some integer k . Then $n^2 + n = (2k + 1)^2 + (2k + 1) = 4k^2 + 4k + 1 + 2k + 1 = 4k^2 + 6k + 2 = 2(2k^2 + 3k + 1)$, which is even. (A1 — same method as Case 1, ft)

Answer Since $n^2 + n$ is even whether n is even or odd, and every integer is one or the other, $n^2 + n$ is even for all integers n . (A1 — conclusion)

EXAMPLE 1.3B (4 MARKS)

Prove that every prime number less than 30 that is 1 more than a multiple of 6 can be written as $6k + 1$ for some positive integer k , and list all such primes.

Step 1 The primes less than 30 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29. (B1 — listing the finite set to check)

Step 2 Check each against the form $6k + 1$: $7 = 6(1) + 1$ ✓; $13 = 6(2) + 1$ ✓; $19 = 6(3) + 1$ ✓. (M1)

Step 3 The remaining primes (2, 3, 5, 11, 17, 23, 29) do not fit the form $6k + 1$ (checking each in turn confirms this). (A1)

Answer So the primes less than 30 of the form $6k + 1$ are exactly 7, 13 and 19. (A1)

EXAM TIP

Check your cases are genuinely exhaustive before you start calculating — “even and odd” is a natural exhaustive split for integers; “positive and negative” is not, because it misses zero. A missed case is one of the most common ways this method loses marks.

1.4 Disproof by counterexample

Concept

A universal claim (“for all $n...$ ”, “every...”) is disproved the moment a single case is found where it fails. This is an important asymmetry: proving a universal claim true requires covering every case (by deduction or exhaustion), but proving it false requires only one case. There is no need to search for a pattern of failures or to explain *why* it fails — exhibiting one counterexample and showing the claim breaks down for it is a complete disproof.

This can feel incomplete compared to the proofs you've just learned — it is tempting to explain more. Reassure them: mathematically, one clean counterexample, correctly checked, earns full marks.

Procedure

1. Read the claim carefully and identify what would make it false.
2. Search for a value that is likely to break the pattern — useful places to look are the edge cases (0, 1, negative numbers) and values that interact awkwardly with the specific numbers in the claim.
3. Substitute that value into the original claim and show clearly that it fails.
4. State the conclusion: this single case is enough to disprove the general claim.

Worked example

EXAMPLE 1.4A (3 MARKS)

Amara claims that $n^2 + n + 1$ is prime for every positive integer n . Show that Amara is wrong.

Step 1 Try small values of n to look for a failure. $n = 1$ gives 3 (prime), $n = 2$ gives 7 (prime), $n = 3$ gives 13 (prime) — these don't help disprove it, so keep trying. Try $n = 4$: $4^2 + 4 + 1 = 21$. (B1 — correct counterexample value) Since $21 = 3 \times 7$, 21 is not prime. (B1 — correct evaluation showing not prime)

Answer Since $n = 4$ is a positive integer for which $n^2 + n + 1$ is not prime, Amara's claim is false. (B1 — explicit statement that one failing case disproves a universal claim)

EXAMPLE 1.4B (3 MARKS)

Disprove the claim: “for all real numbers x , $x^2 \geq x$.”

Step 1 This claim is about all real numbers, so try a value outside the positive integers, such as a fraction between 0 and 1. Let $x = 1/2$. (B1 — sensible choice of counterexample)

Step 2 Then $x^2 = 1/4$, and $1/4 < 1/2$, so $x^2 \geq x$ is false for this value. (B1)

Answer Since $x = 1/2$ is a real number for which the claim fails, the statement “for all real numbers x , $x^2 \geq x$ ” is disproved. (B1)

EXAM TIP

Test 0, 1, negative numbers, and fractions between 0 and 1 as a first reflex whenever you meet a “for all” claim you're asked to challenge — these four categories catch the great majority of exam counterexamples, especially in claims that were only checked against positive integers greater than 1.

1.5 Proof by contradiction

Concept

Proof by contradiction starts by assuming the *opposite* of what you want to prove is true, then reasons forward logically until that assumption leads to something impossible — a contradiction of a known fact, a definition, or itself. Since a true assumption cannot logically lead to something false, the only way out is that the original assumption must have been wrong, which means the statement you wanted to prove is true after all.

This is often the most conceptually strange technique, because it feels backwards: you prove something is true by assuming it's false. A helpful analogy is a courtroom alibi: to prove someone couldn't have committed a crime, you assume they did, follow where that leads, and show it's impossible (they were seen elsewhere at that exact time) — so the assumption of guilt must be false.

This method is essential for statements that can't easily be tackled by direct deduction, most famously proving something *doesn't* exist or *can't* be written a certain way (like $\sqrt{2}$ being irrational).

Procedure

1. **State the assumption:** write clearly “Assume, for contradiction, that [the opposite of the statement] is true.” This sentence is compulsory and often carries its own mark.
2. **Reason forward logically** from that assumption using ordinary algebra and definitions, exactly as in a deduction proof.
3. **Reach a contradiction:** a statement that is clearly false, clashes with the original assumption, or clashes with a known fact (e.g. “a and b share a common factor” when we assumed they had none).
4. **State the conclusion:** because the assumption led to a contradiction, it must be false, so the original statement is true.

Worked example

EXAMPLE 1.5A (5 MARKS)

Prove that there is no smallest positive rational number.

Step 1 Assume, for contradiction, that there is a smallest positive rational number, and call it r . (B1 — clearly stated assumption)

Step 2 Consider the number $r/2$. Since r is positive, $r/2$ is also positive. (M1)

Step 3 Since r is rational, it can be written as a fraction of integers, and $r/2$ is also a ratio of integers, so $r/2$ is rational. (M1)

Answer But $r/2 < r$, since r is positive. So $r/2$ is a positive rational number smaller than r — this contradicts the assumption that r was the smallest positive rational number. (A1 — the contradiction, clearly identified) Since the assumption leads to a contradiction, it must be false. Therefore, there is no smallest positive rational number. (A1 — conclusion)

EXAMPLE 1.5B (6 MARKS)

Prove that $\sqrt{2}$ is irrational.

Step 1 Assume, for contradiction, that $\sqrt{2}$ is rational. Then it can be written as $\sqrt{2} = a/b$, where a and b are integers with no common factor (the fraction is in its simplest form). (B1)

Step 2 Square both sides: $2 = a^2/b^2$, so $a^2 = 2b^2$. (M1)

Step 3 This means a^2 is even (it's 2 times an integer). If a^2 is even, then a itself must be even (an odd number squared is always odd). So write $a = 2c$ for some integer c . (M1 — this step is often rushed; it deserves its own justification)

Step 4 Substitute back: $(2c)^2 = 2b^2$, so $4c^2 = 2b^2$, so $b^2 = 2c^2$. By the same reasoning as before, b^2 is even, so b is even. (A1 — same reasoning as before, ft)

Answer But now both a and b are even, meaning they share a common factor of 2 — this contradicts the assumption that a/b was in its simplest form with no common factor. (A1 — contradiction identified) Therefore the original assumption was false, and $\sqrt{2}$ is irrational. (A1)

EXAM TIP

The step “if a^2 is even then a is even” is doing real logical work and is worth proving separately the first time you meet it (by exhaustion: odd^2 is always odd, so an even square can't come from an odd number). Skipping it is the single most common way this famous proof loses marks.

Chapter summary

Method	Use when	Structure	Key phrase to use
Deduction	Proving a general algebraic identity or property	Represent the general case → manipulate step by step → conclude	“Let n be any integer...”
Exhaustion	The statement covers a small, listable number of cases	Split into cases that cover every possibility → check each → conclude	“Every integer is either even or odd...”
Counterexample	Disproving a “for all” or “every” claim	Find one failing value → substitute and show it fails → conclude	“Let $n = \dots$ then the claim gives ..., which is false”

Method	Use when	Structure	Key phrase to use
Contradiction	Proving something is impossible, or that no such object exists	Assume the opposite → reason forward → reach a contradiction → conclude	“Assume, for contradiction, that...”

Common exam mistakes across this chapter

- **Using specific numbers instead of general letters.** Checking $n = 1, 2, 3$ is never a proof of a “for all” statement — only exhaustion (checking every case in a genuinely finite set) or counterexample (disproving) are allowed to use specific numbers as the actual argument.
- **Missing the concluding sentence.** Even a fully correct algebraic derivation loses the final mark if it doesn't end with a sentence connecting the result back to the original claim.
- **Using the same letter for two independent unknowns.** “Let the two numbers be $2n$ and $2n + 2$ ” only proves a claim about consecutive even numbers, not any two even numbers — a very common way to accidentally narrow a proof.
- **Non-exhaustive cases.** In proof by exhaustion, the chosen cases must genuinely cover every possibility with nothing missed (and ideally no overlap).
- **Manipulating both sides of an identity at once.** When asked to prove an identity, always work from one side towards the other — never cross-multiply or treat it as an equation to solve.
- **Skipping the “assume for contradiction” sentence,** or not clearly identifying what the contradiction actually is at the end of a contradiction proof.

Suggested practice

Once you have learned each technique, try a short mixed set (4–6 short statements) and identify which method to use before attempting a full write-up — recognising the right tool is often the harder skill, and is exactly what exam papers test by not naming the method in the question.

You should now be able to

- Construct a full proof by deduction for a general algebraic statement, ending in a concluding sentence
- Choose the right method — deduction, exhaustion, counterexample, or contradiction — for a given claim
- Write a valid proof by exhaustion using genuinely exhaustive, non-overlapping cases
- Disprove a “for all” claim with a single, correctly checked counterexample
- Write a proof by contradiction, including the classic proof that $\sqrt{2}$ is irrational